

Training ESM to Predict Viable Flu Strains

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Why Use ESM for Flu Prediction?

- Influenza is constantly evolving, each year public health organizations and researchers must develop a new vaccine.
- To produce an effective vaccine, forecasting models are used to predict future viral variants.
- ESM and other large language models may be an effective tool to assist in predicting next season's flu strains more accurately, by uncovering new patterns in the proteins of successful flu variants.

Introduction

ESM

Evolutionary Scale Modeling (ESM) is a large language model developed by Meta, which can predict protein viability, structure and function. In this project ESM2 was used, but will be referred to as ESM in this poster.

PRETRAINING

Machine learning models are fed huge amounts of data to infer associations, for ESM the largest model was trained on 15 billion proteins from uniparc.

FINE-TUNING

These models can be further refined and trained on specific data.

MKTHIALSYIFCL*LGQD*PGNDNSTATLC*GHHAVPNGTLVKT*TDDQIEVTNATEL*QS

MKTHIALSYIFCLALGQDLPGNDNSTATLC LGHHAVPNGTLVKTITDDQIEVTNATELVQS

ESM is fed sequences of interest, where amino acids are randomly masked, and ESM is tasked to predict the missing amino acids.

Nextstrain

Nextstrain is an interactive tool developed by the Bedford and Neher lab, that is used to generate interactive phylogenetic trees from pathogen data.

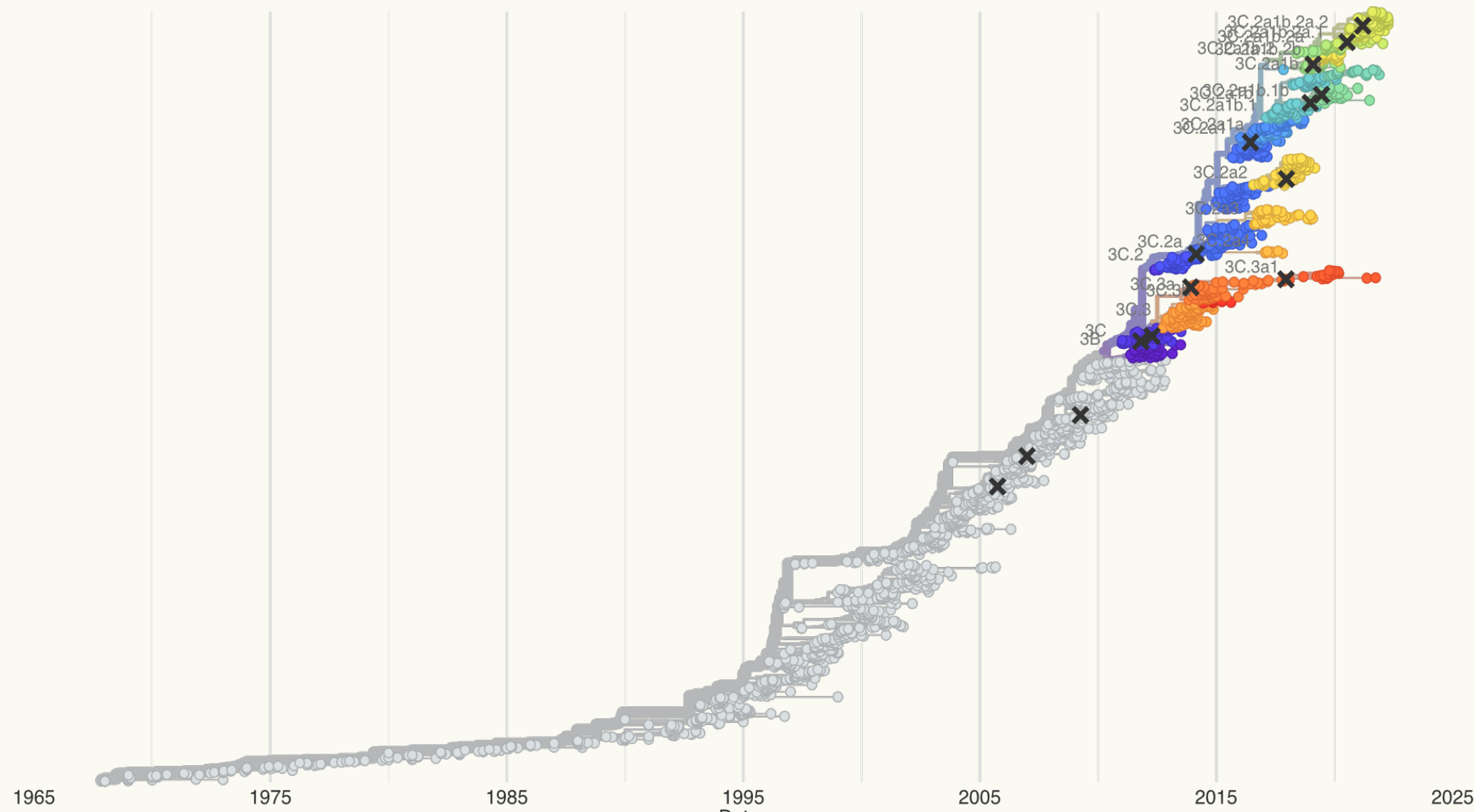
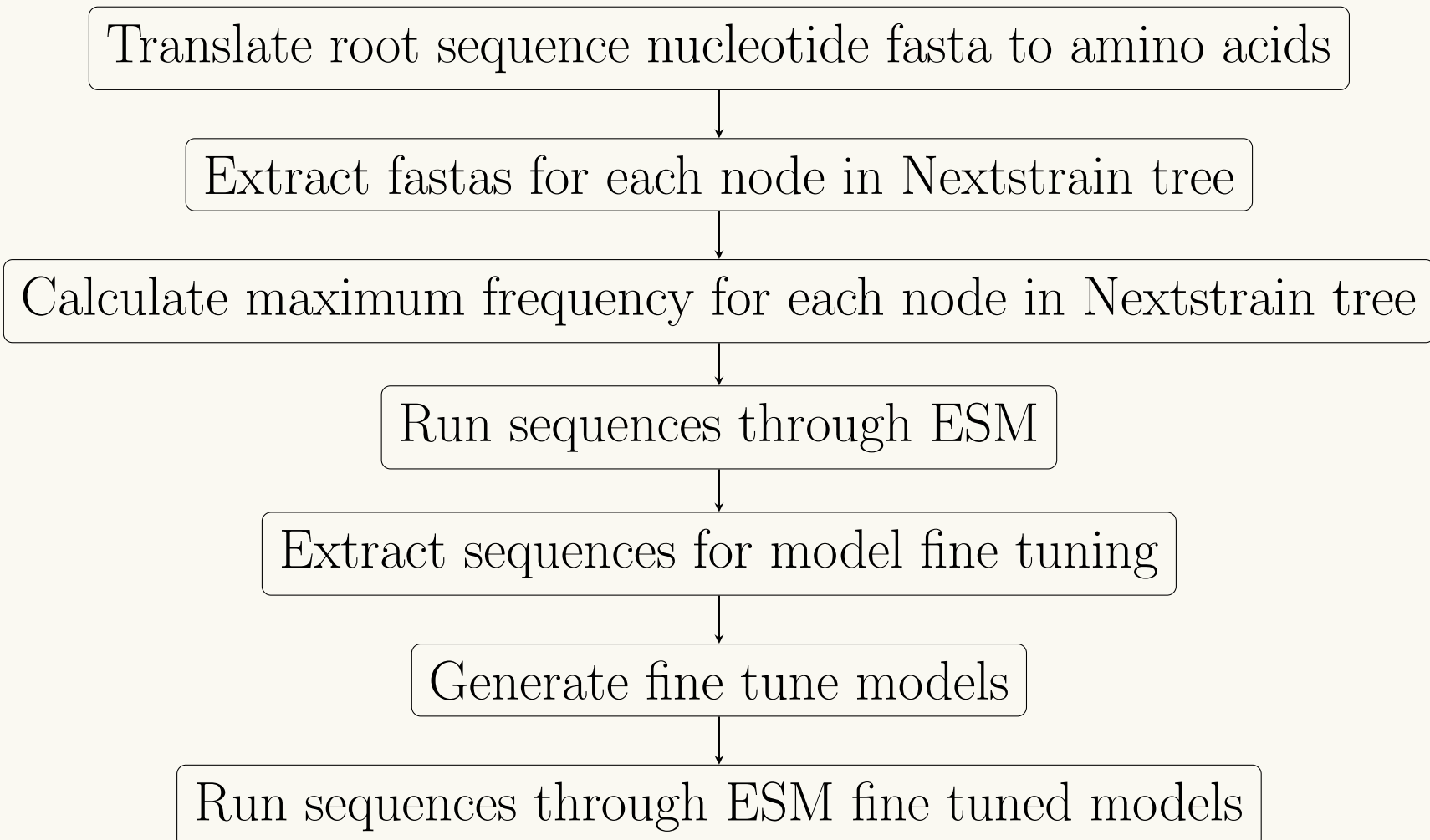


Figure 1: 60 year Nextstrain Tree for H3N2, for flu segment HA. Each dot is a node on a tree. The x-axis depicts time, with more recent, more fit variants further the right of the figure. Nodes are colored here by clade, but can be colored by any metadata information with Nextstrain.

Pipeline



Snakemake pipeline workflow: a pipeline was written to run multiple processes in parallel on Fred Hutch's Conatus server, a high performance computing GPU cluster. The pipeline extracts fastas for each node in nextstrain tree, to run them through ESM models. The pipeline also fine tunes ESM models and runs fastas through the fine tuned models.

Running All Three Base ESM Models

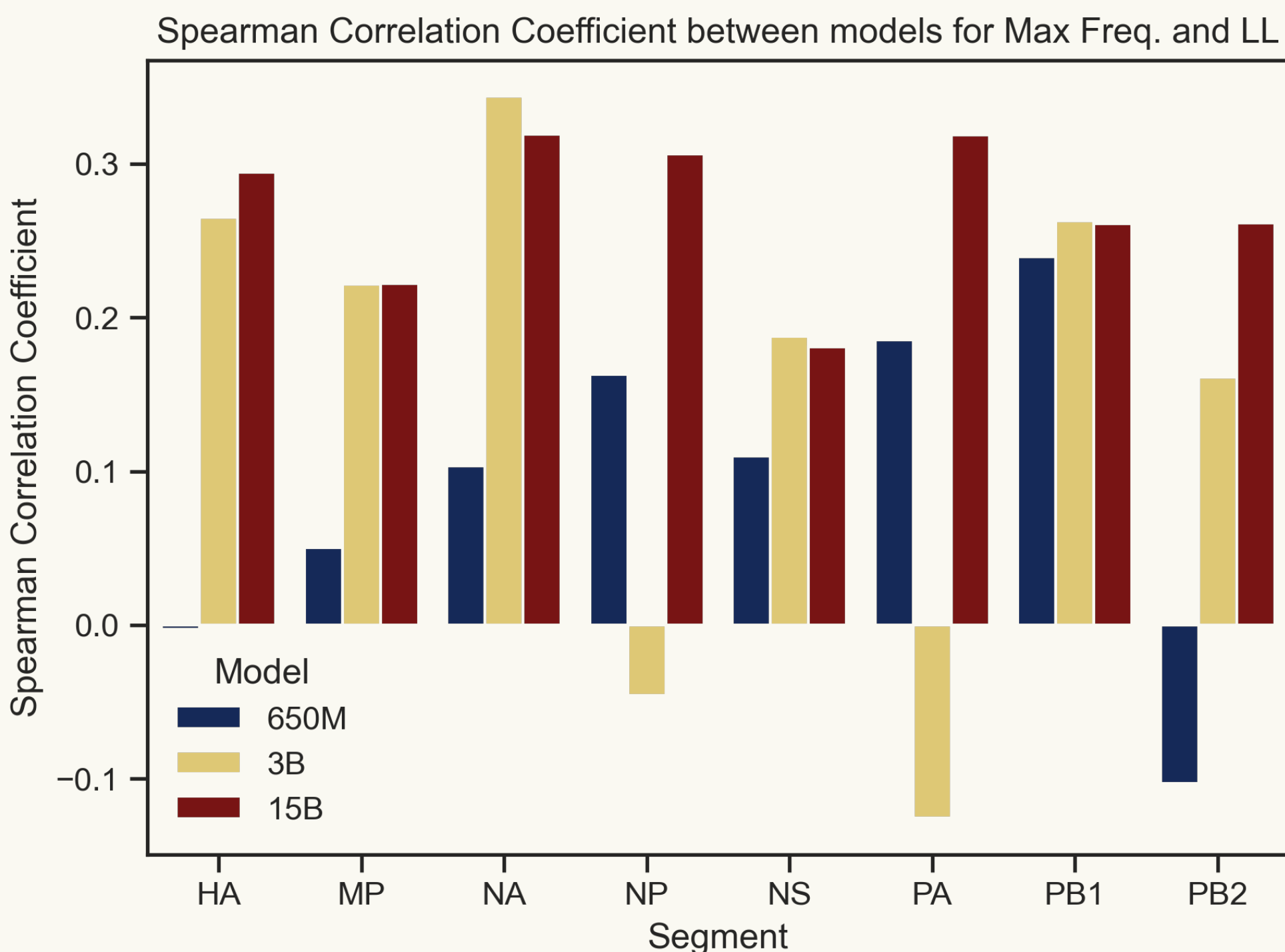


Figure 2: Spearman correlation coefficient (cc) for ESM log likelihood vs maximum frequency from Nextstrain trees for base models trained on 650 million proteins, 3 billion proteins and 15 billion proteins. Each ESM model was run separately for each flu protein. Models with more proteins produced predictions more similar to Nexstrain maximum frequency (Higher spearman cc).

Fine Tune vs Base 3B ESM Model Log Likelihood Over Time

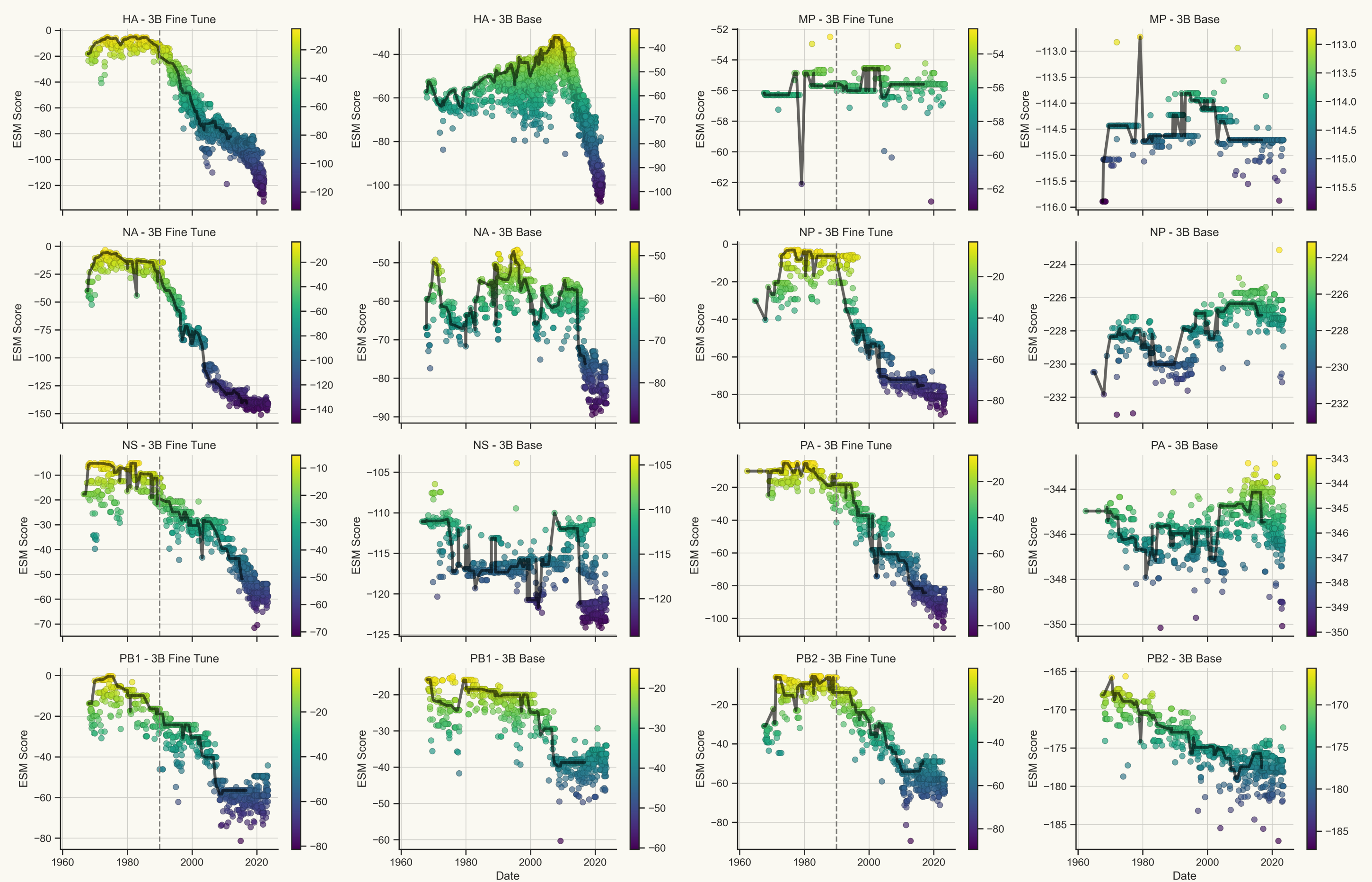


Figure 3: ESM log likelihood vs time, for both fine tune and base 3B models. For fine tune models ESM score drops off more linearly over time the further from the original training data. Dashed grey line shows cutoff for training data. Dark grey line (trunk), connects nodes where maximum frequency is 100%. The trunk (the main evolutionary lineage) shows us if branches fall above or below trunks. Branches or side lineages, represented just as nodes here.

Comparing Fine Tune Models to Base Models

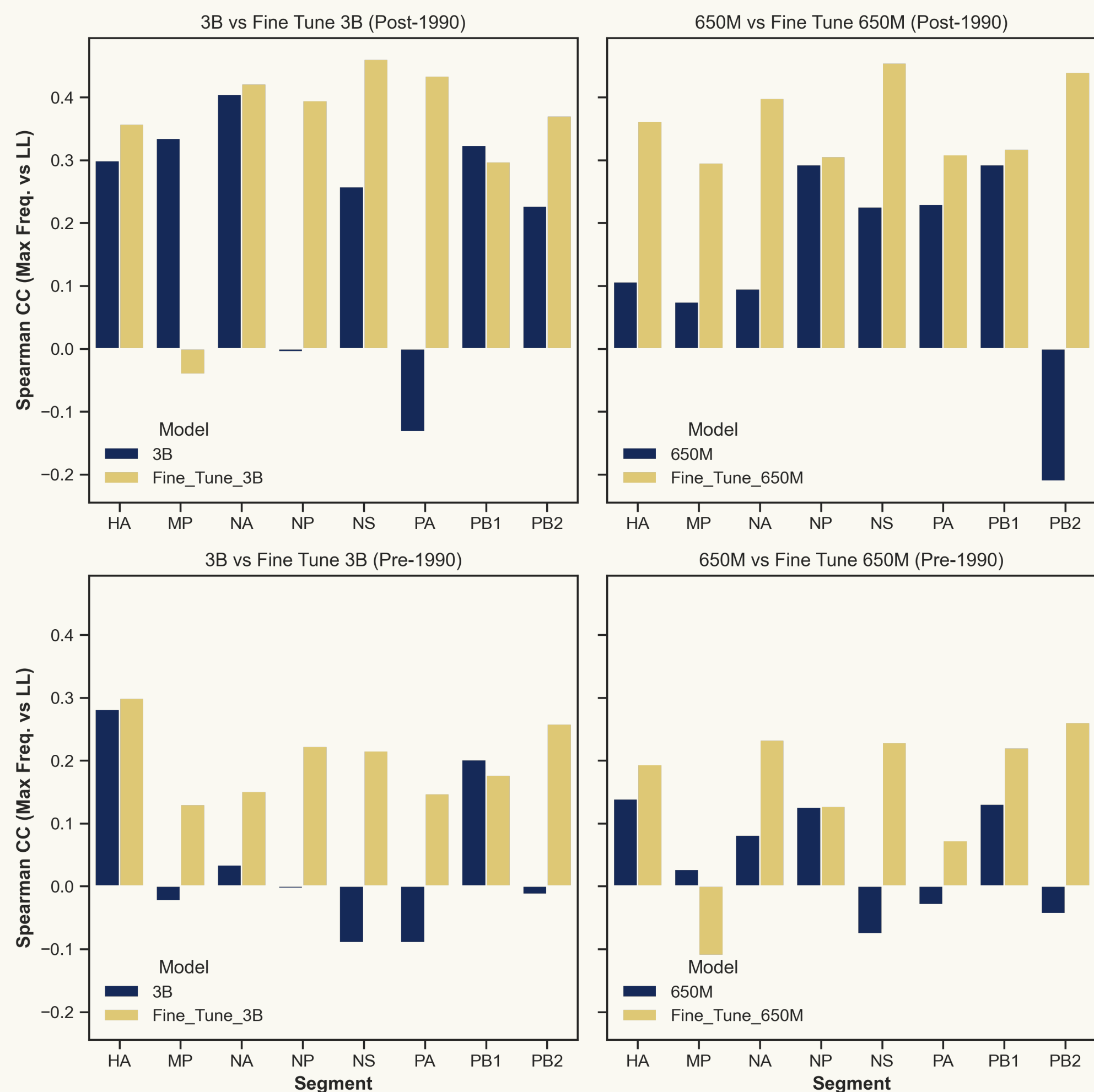


Figure 4: 650M and 3B base models compared to fine tune models, trained on data from Nextstrain trees before 1990. Spearman cc (ESM log likelihood vs maximum frequency) for both pre-1990 (training data) and post-1990 (test data) for each flu segment. Spearman cc is generally higher for the larger protein model (3B) and for fine tune models across all models. Spearman cc is lower than expected for fine tune models on runs with training data (pre-1990).

Adjusting Fine Tuning Parameters

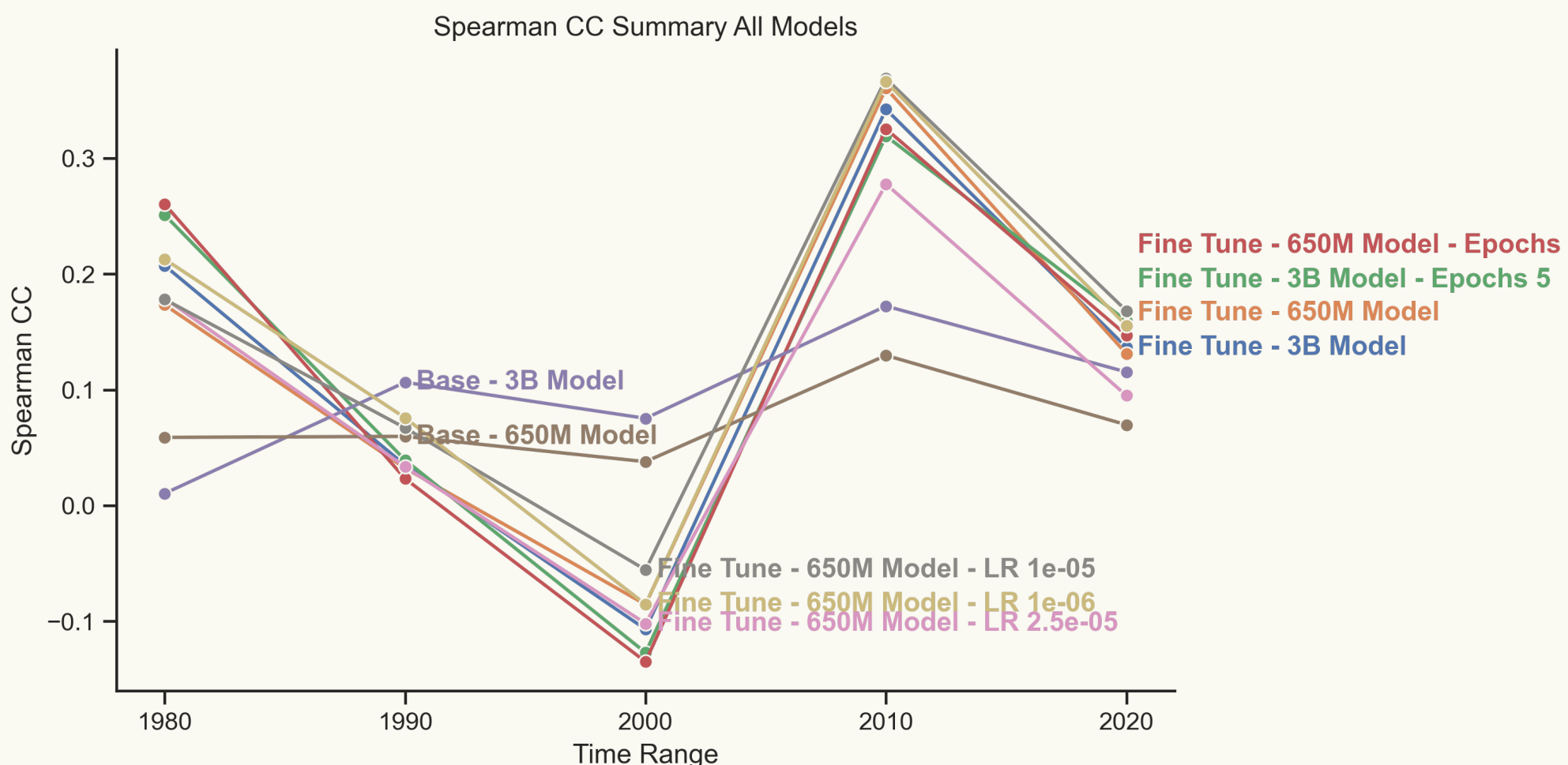


Figure 5: Spearman cc (ESM log likelihood vs maximum frequency) comparing base models, and fine tune models with adjustments to learning rate and number of epochs. Each time point is a test run on a 20 year window. Regardless of tweaking there is over-training present across all fine tune models with a large spike at 2010. Log likelihood would be expected to decrease the further in time sequences are from the training data, the spike is suspected to be mutations reappearing which were also present in the training data. Fine tune models were trained on data before 1990 from the Nextstrain tree.

Next Steps

- Compare the ability of ESM vs DMS to forecast evolutionarily successful influenza viruses.
- Train models on larger amounts of sequences to prevent over-training in fine tuning and run ESM on other viruses: SARS-CoV-2.



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